

Lab #05: SYSTICK & GPIO

Overview

You will write the ARM Assembly code to make the built-in LEDs on the LaunchPad blink with different timing **using the SysTick timer to precisely time the blinks**. You **MUST** use the **SysTick timer** for the square wave timing. The colors and timings are controlled by the built-in switches.

Note:

- Use **Port F** on the LaunchPad for this lab.
- A **starter project** has been provided for you.

TASK 1 – Simulation Demonstration (70 PTS)

You must program your LaunchPad to behave in the following way.

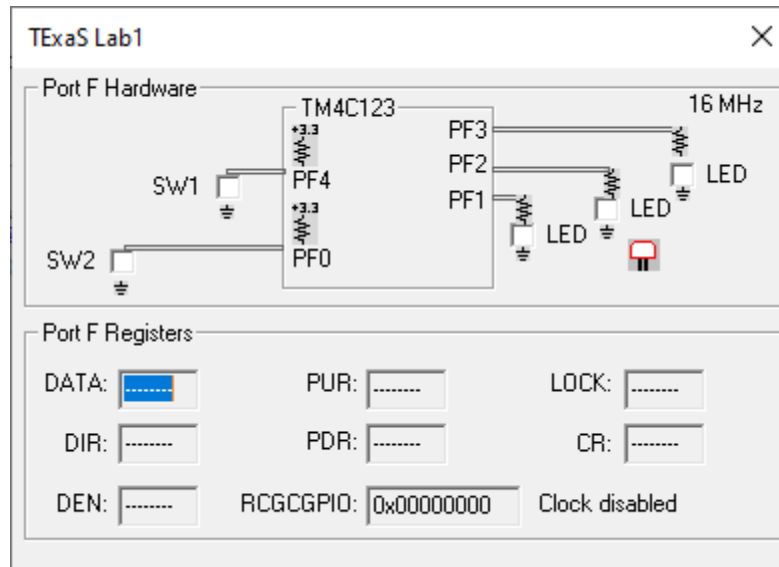
- All LEDs on the LaunchPad will be off at start-up and until any buttons are pressed.
- The first time SW2 is pressed...
 - The red LED will blink with 50/50 duty cycle and period of ~0.75 s.
- The second time SW2 is pressed...
 - The blue LED will blink with 50/50 duty cycle and period of ~0.5 s.
- The third time SW2 is pressed...
 - The green LED will blink with 50/50 duty cycle and period of ~0.25 s.
- All subsequent presses of SW2 will repeat the above cycle.

You can think of it as a state machine where state 0 (S0) is the idle state (all LEDs off) and then each press cycles through S1 (red blink) → S2 (blue blink) → S3 (green blink) → S1 → S2 → ...

If SW1 is pressed, the board should go back to the idle state (S0) and wait for another SW2 press.

You will demonstrate on the simulator first to show proper simulation. For debugging, you can set the period shorter. Once you have proper behavior, then change the timing to the behavior described above. We don't have a way to do precise timing yet, so these times are approximate.

Do not worry about the case when both buttons are pressed. We will assume that never happens.



TASK 2 – Hardware Demonstration (30 PTS)

Download your code to the LaunchPad and demonstrate that your code works correctly on the LaunchPad using the built-in LEDs and switches (SW1 & SW2).

You will most likely have to change your code to tweak the timing on the actual board.

Submission (Through Canvas, BOTH Partners)

Submit the complete and working project to Canvas as one zip file.

- The project file will be named Lab5.uvprojx
- The project folder will be named ECE_3436_Lab5_Team##.
- Therefore, the zip file will be named ECE_3436_Lab5_Team##.zip.

Meeting/In-person Demonstration

If possible, please demonstrate your code on the hardware to me in-person or through a video meeting. If you choose not to demonstrate, we will grade based only on your code. That means, if the code doesn't run well on our hardware, you will lose your points.